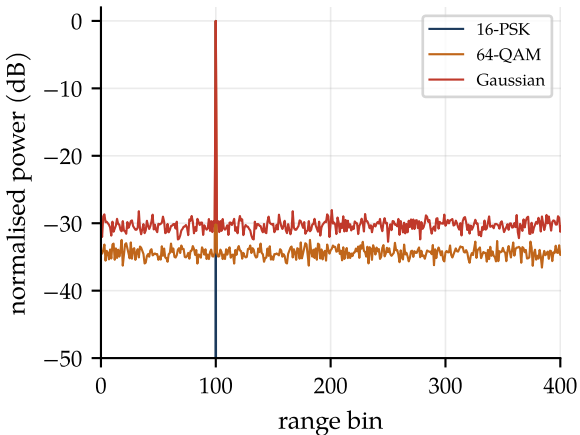


(a) correlation receiver, $N = 1024$



(b) the deterministic--random trade-off

